Voltage regulator

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Cuprins

1.Project requirement

Design a voltage regulator using the following specification: input voltage Vi=220V, minumum and maximum output voltage [Vo1,Vo2] = [4;16] and the output current limitation Iolim[A] =2.5.

2.Theoretical fundamentation

2.1 Block diagram

For a better understanding of the circuit, I made a block diagram, which clearly explains the functionality of the circuit and the flux of the signal.

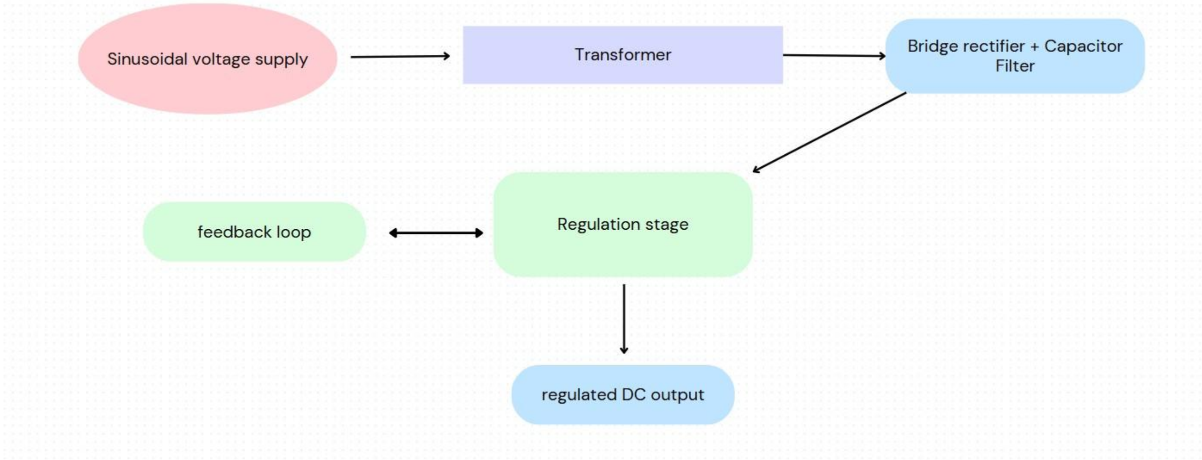


Image 1: the block diagram of the voltage regulator

From Image 1, we can easily analize the functionality of the linear voltage regulator circuit. The power supply of the circuit is a sinusoidal voltage of 311V (peak value), with a frequency of 50Hz. With the help of the transformator, this voltage steps down to the value of 24V (effective value). Both, the bridge rectifier and the capacitor filter, convert the AC voltage from the transformer's secondary into a DC voltage and smooth out the ripples. After this process, we obtained an unregulated voltage. Here is the time where the regulation stage appears. Consisting of a zener diode that provides a stable voltage, an operational amplifier that acts like a comparator and two transistors, this part of the circuit does the main procedure of regulating the implemented voltage, resulting a regulated DC output. Also, we can not forget about another important factor : the feedback loop where a portion of the DC output voltage is fed back to the regulation stage (specifically to the error amplifier) to allow the circuit to self-adjust and maintain the desired output voltage.

2.2 The Transformer

This part of the circuit performs AC voltage step-down from a high peak value to a lower peak (and RMS) value. The values of the inductances determine the turns ratio and thus the voltage transformation ratio. The resistors represent source impedance, a minimal load (R1), and the actual load (R2) where the stepped-down voltage is delivered.

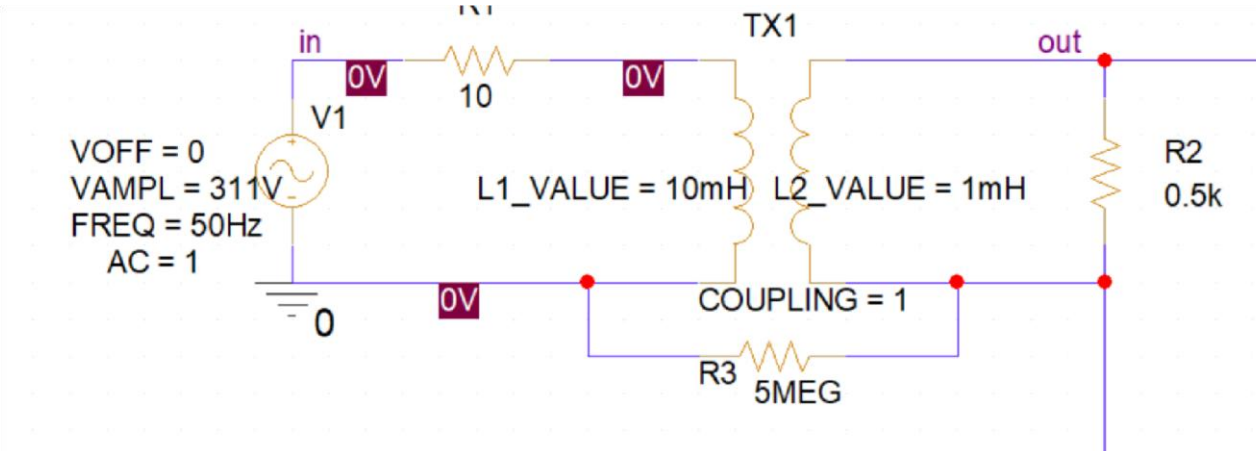


Image 2:Electrical schematic of the transformer

In the Image 2 we can see that we choose an amplitude of 311V, which represents the peak value of 220V which is the effective (Rood Mean Square) value. For this calculation we used the formula:

$$Vpeak = Vrms * \sqrt{2} \qquad (1)$$

Where the factor $\sqrt{2}$ comes from varies equations, as the following:

1. Definition of RMS value

$$Xrms = \sqrt{\frac{1}{T}\int_0^T [x(t)]^2} \qquad (2)$$

2. Mathematical representation of a sinusoidal wave

$$v(t) = Vpeak * \sin(wt) \qquad (3)$$

3. Applying the RMS definition to a Sine Wave:

$$Vrms = \sqrt{\frac{1}{T}\int_0^T [Vpeak * \sin(wt)]^2 dt} \qquad (4)$$

$$Vrms = \sqrt{\frac{1}{T}\int_0^T Vpeak^2 * sin^2(wt) dt} \qquad (5)$$

Since *Vpeak* is a constant, we can take it aut of the integral:

$$Vrms = Vpeak\sqrt{\frac{1}{T}\int_0^T sin^2(wt)dt} \quad (6)$$

We need to evaluate the integral of $sin^2(wt)$ over one period. We can use the trigonometric identity:

$$sin^2(x) = \frac{1-\cos(2x)}{2} \quad (7)$$

Therefore, the factor of $\sqrt{2}$ arises from the mathematical process of finding the equivalent DC value of a sinusoidal waveform that delivers the same power to a resistor, which involves squaring the sine function, averaging it over a cycle, and then taking the square root.

If we were to talk about how do you calculate the values of the coils from the transistor, we have to keep in mind that the voltage ratio of a transformer is directly proportional to the turns ratio:

$$\frac{Np}{Ns} = \frac{Vp}{Vs} \quad (8)$$

Where Np is the number of turns in the primary coil, Ns is the number of turns in the secondary coil, Vp is the RMS voltage across the primary coil (220V) and Vs is the RMS voltage across the secondary coil (desired output, 15V).

We use the relationship between the voltages and inductances in an ideal transformer:

$$\frac{Vsec\ rms}{Vprim\ rms} = \sqrt{\frac{L2}{L1}} \quad (9)$$

The resistor R3 has a significant high value because it represents a very high impedance load. In some simulations, a large resistor like this might be used to approximate an open circuit or to provide a minimal load for voltage measurement.

We can view the change of the voltage amplitude by simulating a Time Domain analysis.

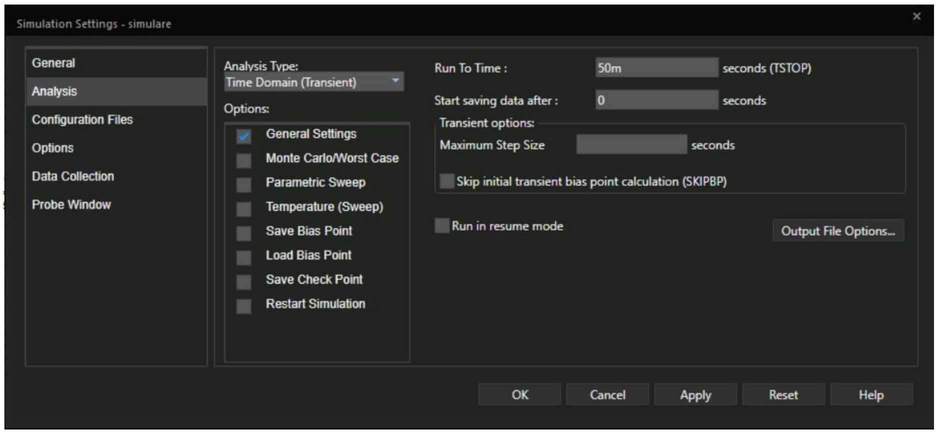


Image 3: Simulation settings

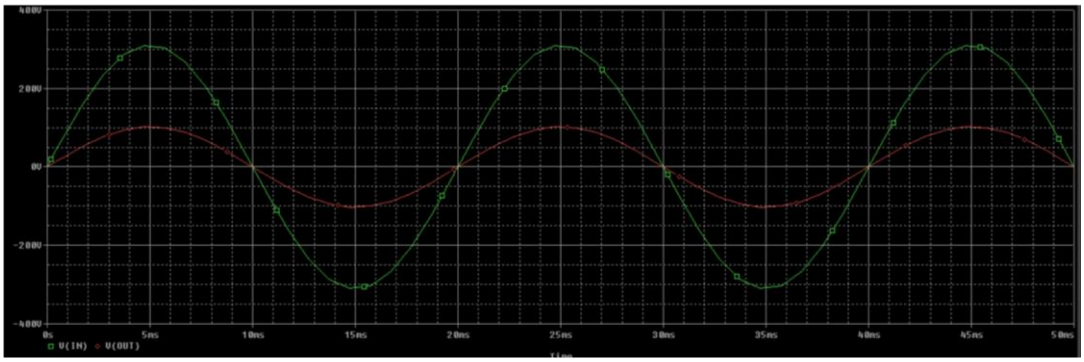


Image 4: Time Domain Analysis

2.3 Bridge Rectifier and the Capacitor Filter

The diode bridge forms a full wave rectifier configuration that coverts alternative current voltage into a pulsating direct current voltage. During one half-cycle of the AC input, two diodes conduct, allowing current to flow in one direction through the load (which would be connected to the "rectif" output). During the other half-cycle, the other two diodes conduct, again forcing the current to flow in the same direction through the load. This results in a DC voltage that rises and falls with the absolute value of the AC input.

The filter capacitor smooths out the pulsating DC voltage from the rectifier, reducing the ripple and producing a more stable DC voltage. The capacitor charges up to the peak voltage of the rectified waveform. When the rectified voltage drops (between the peaks), the capacitor uses the resistor R4 to discharge, supplying current to the load and maintaining the voltage level to some extent. The larger the capacitance, the better the filtering and the lower the ripple.

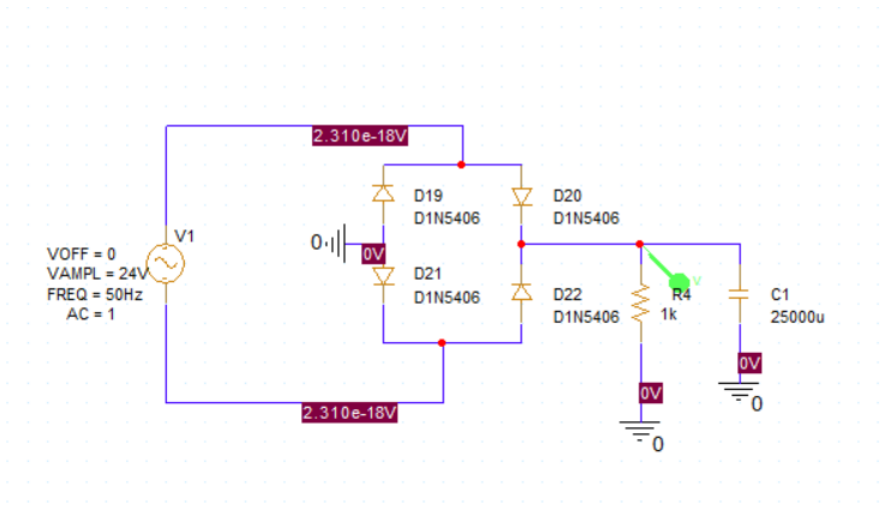


Image 5: Electrical scheme of the Bridge rectifier and capacitor filter

I chose D1N5406 for the diodes because they accept current up to 3V and for the value of the capacitor we have the following formula:

$$C = \frac{I}{f\Delta V} \tag{10}$$

Where I=2.5A, f=100Hz for a full-wave rectifier and $\Delta V = 1V$ ripple.

If we want see the functionality of this part, we can run in Pspice a Bias Point simulation, which generates the values on the *Image 6*, but also we can run a Transient Time simulation, where we can observe better the rectification of the signal.

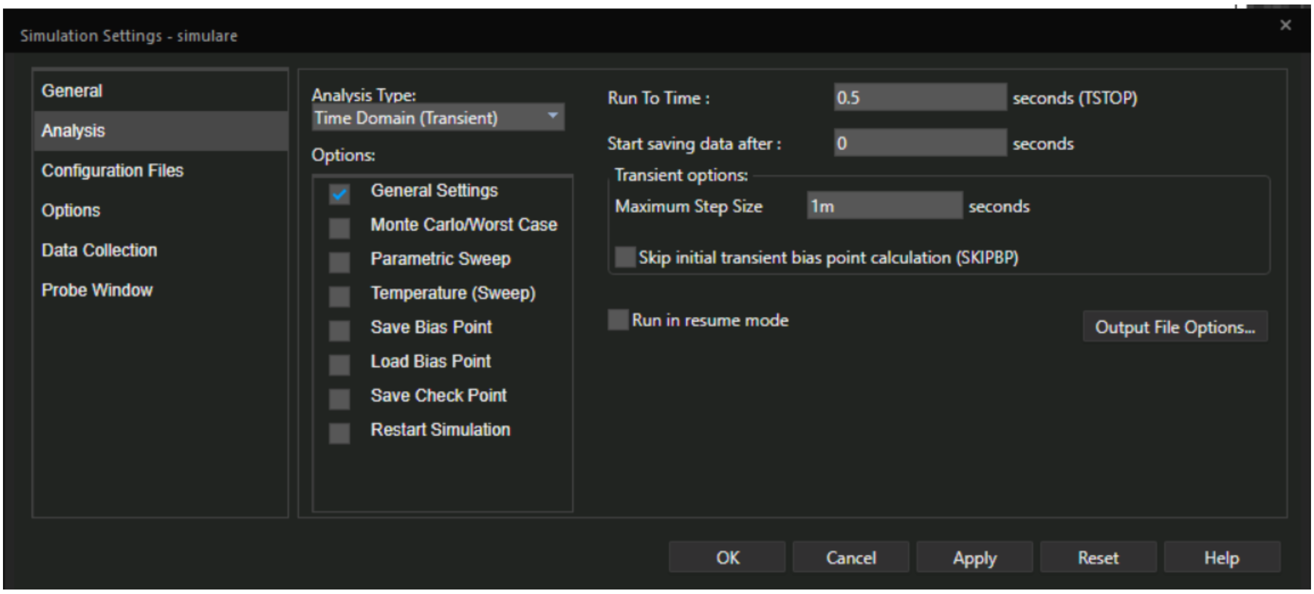


Image 6: Simulation settings

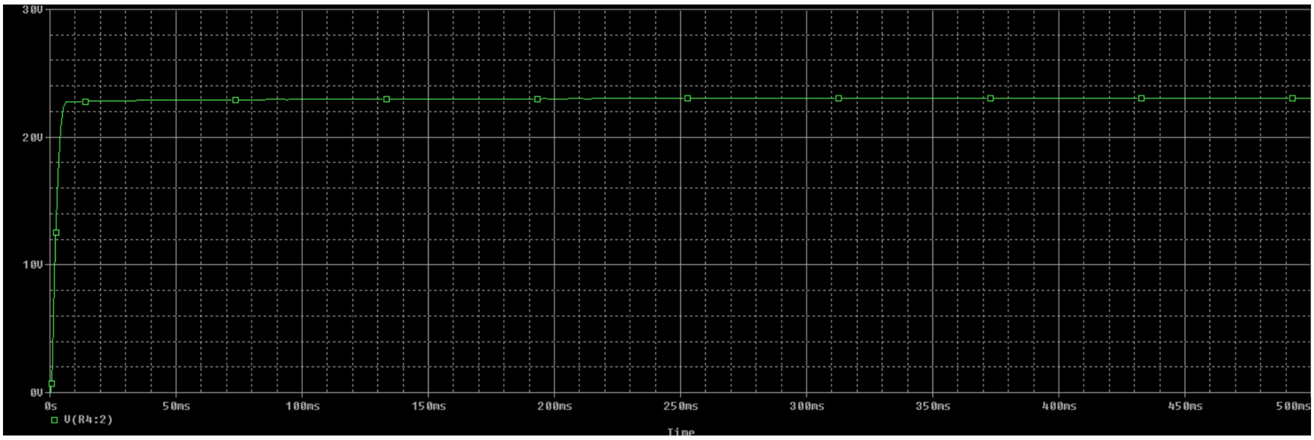


Image 7: Rectifier simulation results

As we can see, the rectifier draws the transformer's AC voltage and converts it into a stable DC output, smoothed out by the capacitor, maintaining around 24V.

The high voltage suggests that the circuit is likely intended to regulate the current to a specific limit, and the output voltage will depend on the load resistance.

2.4 Regulation stage

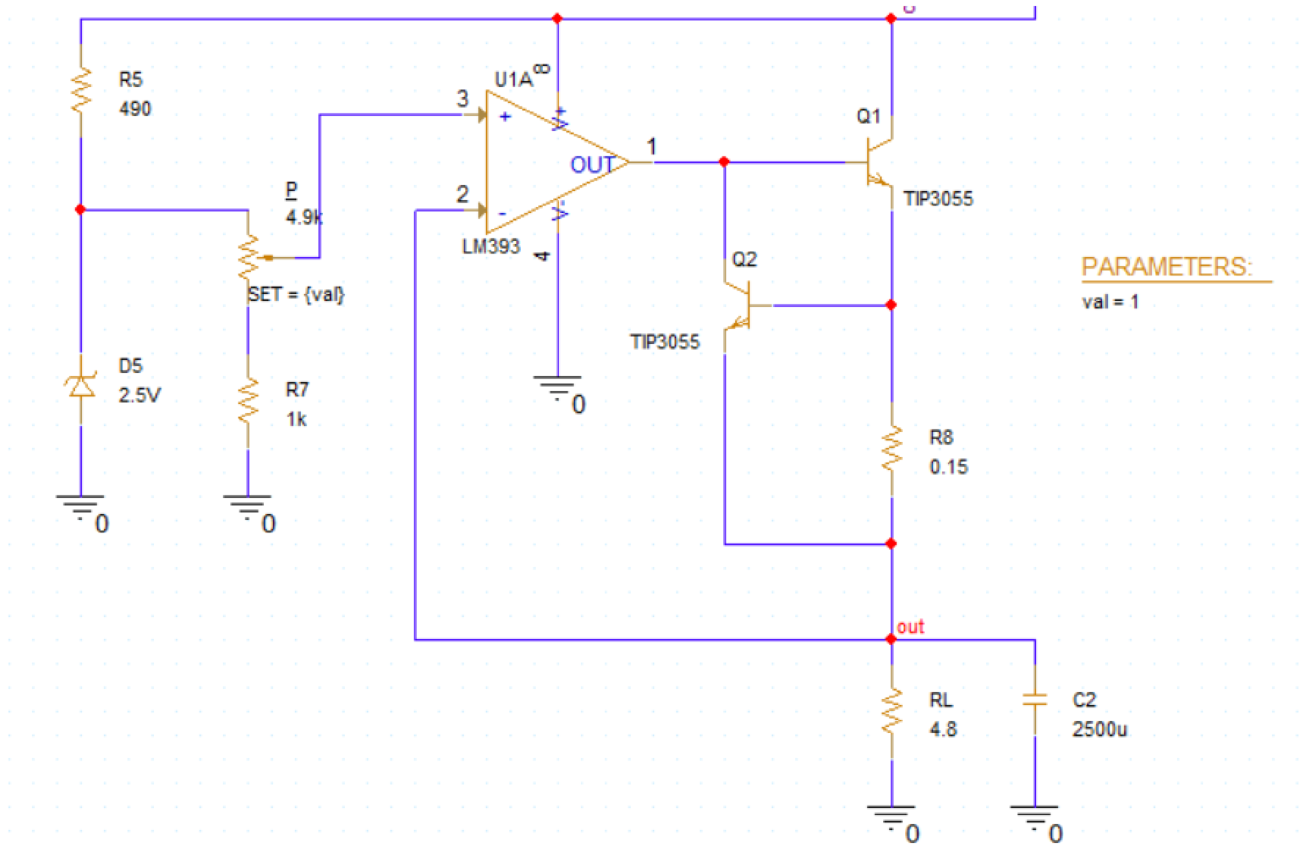


Image 8: Linear Voltage Regulator

The regulation part of the circuit is responsible for maintaining a stable output voltage despite variations in the input voltage or the load current. Here's a breakdown of how each component contributes to this regulation:

1. Zener Diode D5 (2.5V) and Resistor R5 (490Ω): The Voltage Reference

The Zener diode D5 provides a stable reference voltage. When a reverse voltage is applied to it (above its Zener voltage), it maintains a relatively constant voltage across its terminals, regardless of small changes in current.

Resistor R5 limits the current flowing through the Zener diode. It ensures that the diode operates within its specified current range. The voltage across D5 (approximately 2.5V) serves as the reference voltage (Vref) for the regulation process.

2. Resistors P (4.9kΩ Potentiometer) and R7 (1kΩ): The Feedback Network (Voltage Divider)

These resistors form a voltage divider connected across the output voltage. The voltage at the junction of P and R7 is a fraction of the output voltage (Vout). This fraction is fed back to the inverting input (pin 2) of the operational amplifier.

By adjusting the potentiometer, you change the ratio of the voltage divider. This allows you to set the desired output voltage. A higher resistance for the upper part of R6 will result in a larger fraction of Vout being fed back, and the regulator will adjust to maintain a higher overall Vout.

R7 provides a fixed resistance in the divider. Its value, in relation to P, determines the range of output voltages that can be achieved.

4. Operational Amplifier U2A (LM393): The Error Amplifier

The op-amp acts as an error amplifier. It compares the voltage at its non-inverting input (pin 3) with the voltage at its inverting input (pin 2).

Inputs: Non-inverting input (pin 3): Connected to the Zener reference voltage (Vref).

Inverting input (pin 2): Connected to the feedback voltage from the voltage divider (a fraction of Vout).

Output (pin 1): The op-amp amplifies the difference between its two input voltages. If the feedback voltage (from the divider) is lower than the reference voltage, the op-amp's output voltage will increase. Conversely, if the feedback voltage is higher than the reference, the opamp's output voltage will decrease. This amplified error signal drives the transistors to adjust the output voltage.

5. Transistors Q1 and Q2 (TIP3055): The Pass Elements

These transistors are the pass elements in this linear regulator. They act like variable resistors controlled by the output of the op-amp. They are in series with the unregulated input voltage and the load.

The op-amp's output voltage controls the base current of Q1 (and subsequently Q2).If the opamp's output voltage increases, the base current of Q1 increases, causing it (and Q2 in a Darlington configuration) to conduct more current. This leads to a higher voltage across the load (Vout).

If the op-amp's output voltage decreases, the base current decreases, causing the transistors to conduct less current, resulting in a lower Vout .

Darlington Configuration (Q1 and Q2): Connecting Q1 and Q2 in a Darlington configuration provides a higher overall current gain. This means that a small change in the op-amp's output voltage can control a larger current flow to the load.

6. Resistor R8 (0.15Ω): Current Sensing Resistor

R8 is a small-value resistor placed in series with the output current. The voltage drop across R8 is proportional to the output current ($VR8 = Iout \times R8$).

Role in Regulation (Likely for Current Limiting): While not directly involved in the voltage regulation feedback loop shown going to the op-amp, the voltage across R8 is likely used by another part of the circuit (not fully visible) for current limiting or overcurrent protection. If the output current exceeds a certain limit, the voltage across R8 will become high enough to trigger a mechanism that reduces the current flow, protecting the transistors and the load.

How Regulation Occurs:

Setting the Output Voltage: The desired output voltage is set by adjusting the potentiometer P, which changes the feedback ratio.

Maintaining Stability: If the output voltage starts to drop (due to an increase in load current or a decrease in input voltage):

The feedback voltage at the inverting input of the op-amp (pin 2) decreases.

The op-amp sees a larger difference between its non-inverting and inverting inputs, and its output voltage increases.

The increased op-amp output voltage increases the base current of the transistors Q1 and Q2, causing them to conduct more.

This increased conduction allows more current to flow to the load, bringing the output voltage back up to the desired level.

Correcting Overvoltage: If the output voltage starts to rise:

The feedback voltage at the inverting input of the op-amp (pin 2) increases.

The op-amp sees a smaller difference (or a reversed difference) between its inputs, and its output voltage decreases.

The decreased op-amp output voltage reduces the base current of the transistors, causing them to conduct less.

This reduced conduction decreases the current flow to the load, bringing the output voltage back down to the desired level.

In essence, this circuit uses negative feedback to continuously monitor the output voltage and make adjustments through the pass transistors to keep it stable at the value determined by the reference voltage and the feedback network. The op-amp acts as the control element, sensing the error and amplifying it to drive the correction mechanism.

2.5 Calculation of the resistances

In order to size our resistances, including the potentiometer, we will consider our Zener diode 2.5V and R7 as 1k. Following the formula:

$$Vout = Vz * (1 + \frac{Rtop}{R7}) \qquad (11)$$

If we replace Vout with 16V, Vz and R7, we obtain for *Rtop* a value of 5,4k ohms. Knowing that :

$$R5 + P = 5{,}4k \qquad (12)$$

We can estimate that R5 is around 490 ohms, remaining for P the value of 4.9k.

Dividing Vout with Iout, we get a range of values for RL between [1,6 ; 6,4] OHMS. I choose a value between those two, so there we have 4,8 ohms.

$$RL = \frac{Vout}{Iout} \qquad (13)$$

Our OpAmp could shut off the pass transistor if Vr8 would be higher than V threshold, which is typically 0.6V for BJTs, and considering our Ilimit is 2.5A we get:

$$R8 = Ilimit * 0.6 = 1.5 \qquad (14)$$

3.Circuit implementation

The circuit is implemented based on the components described and analyzed in the theoretical foundation part.

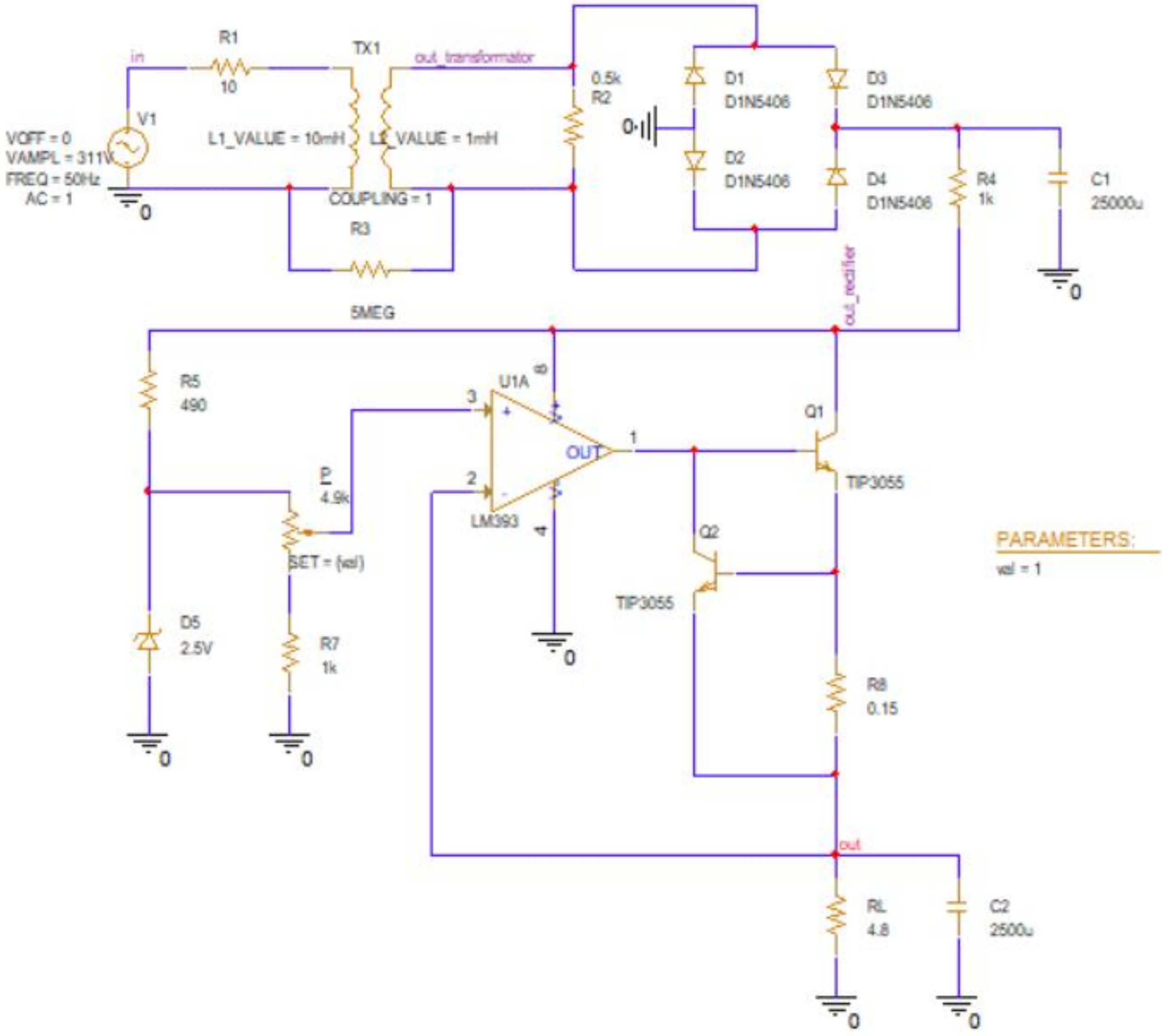


Image 9: Electrical scheme for the voltage regulator

4.Simulation results and Statistical Analysis

4.1 Simulation results

With our full schematic ready we can begin running our simulation and notice the results. Our first simulation will be a transient analysis as we are interested in seeing the behaviour of the circuit trough time. We will also be doing DC Sweep analysis as we are working with a DC signal being fed to our regulator’s OpAmp.

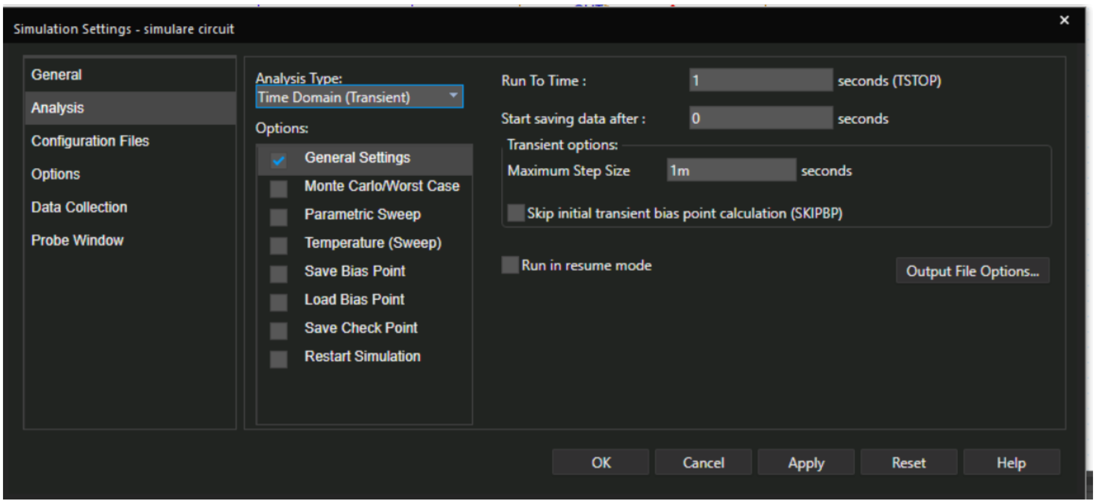


Image 10: Voltage Regulator Simulation Settings

Our first simulation is the TIME DOMAIN (TRANSIENT) ANALYSIS

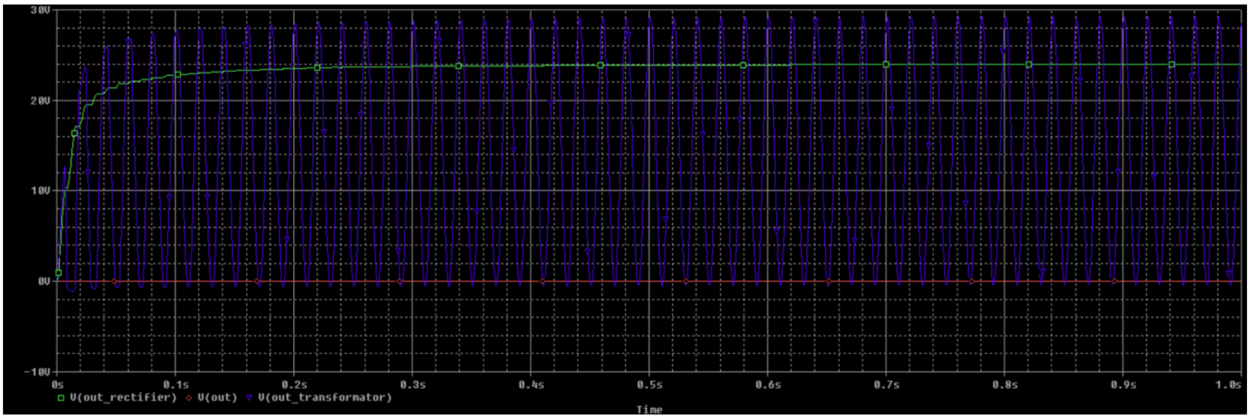


Image 11: TIME DOMAIN (TRANSIENT) ANALYSIS with P=0

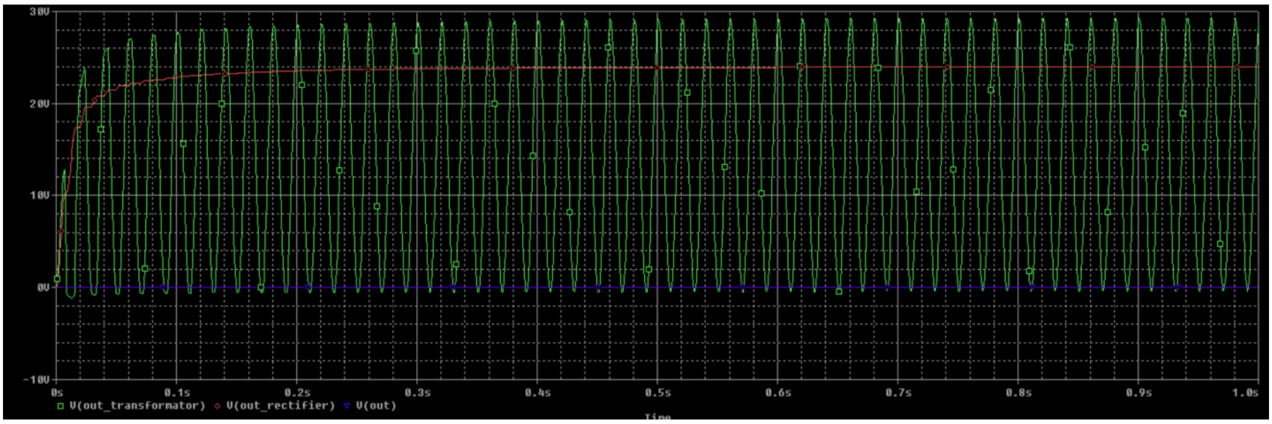


Image 12: TIME DOMAIN (TRANSIENT) ANALYSIS with P=1

For the next simulation, we will try a DC Sweep analysis with the potentiometer as a global parameter with the start and end value as our output limits

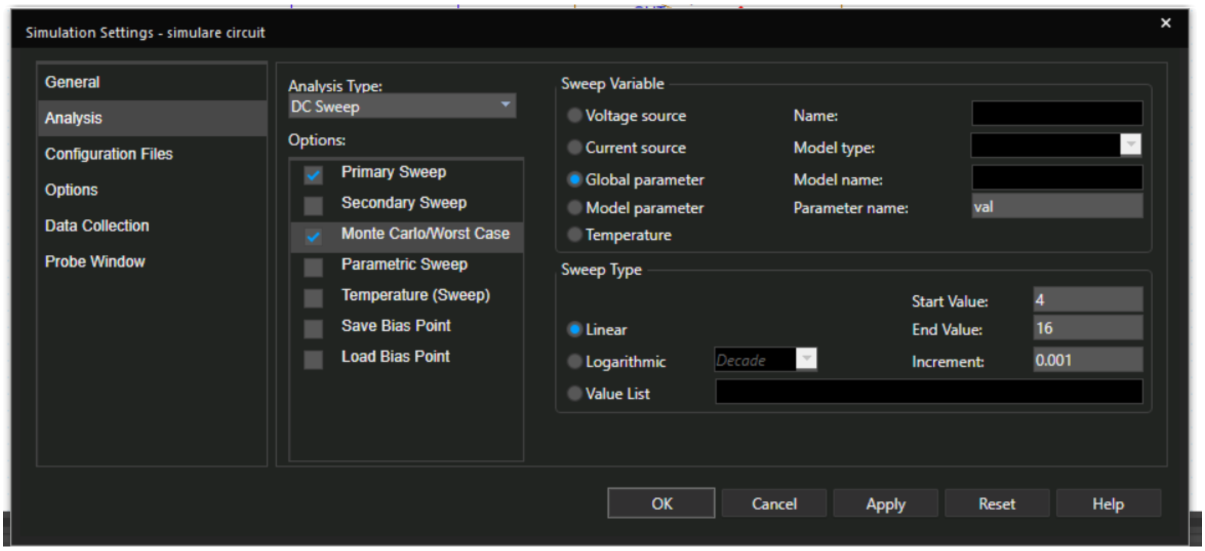


Image 13: DC SWEEP settings

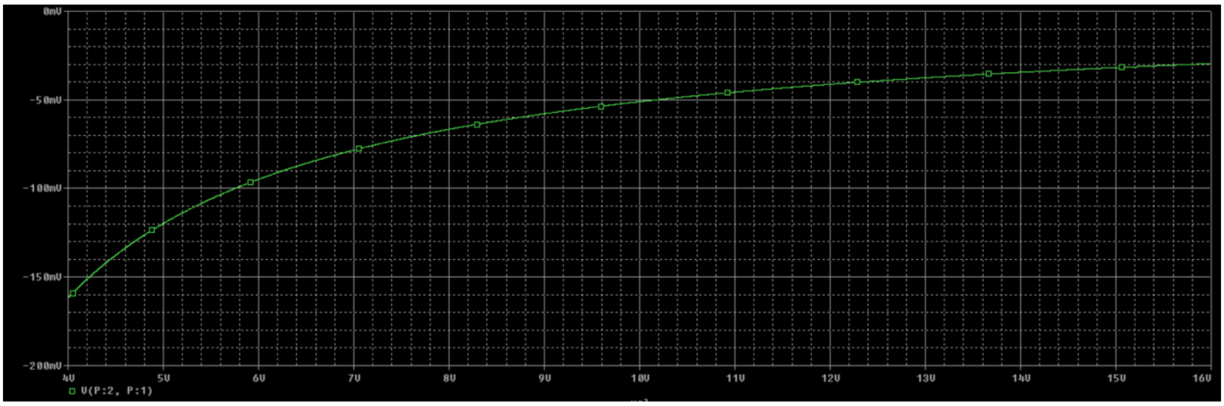


Image 14: DC SWEEP ANALYSIS

4.2 Statistical Analysis

For both, Monte Carlo and Worst Case, we study the dissipated power by RL.

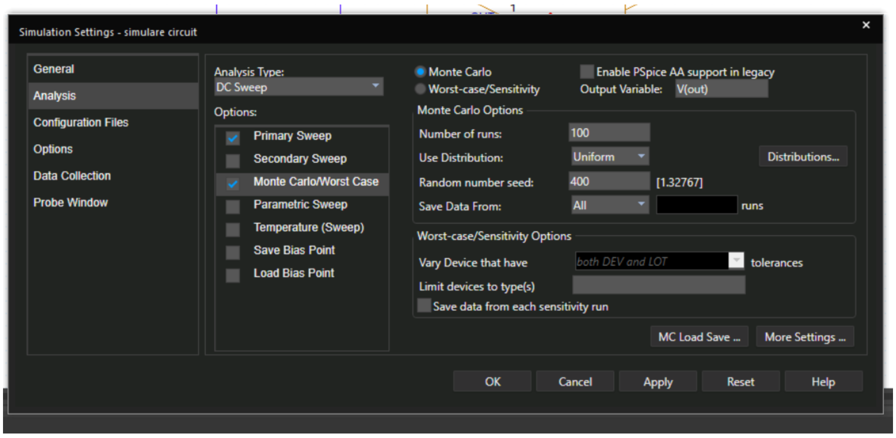


Image 15: MONTE CARLO settings

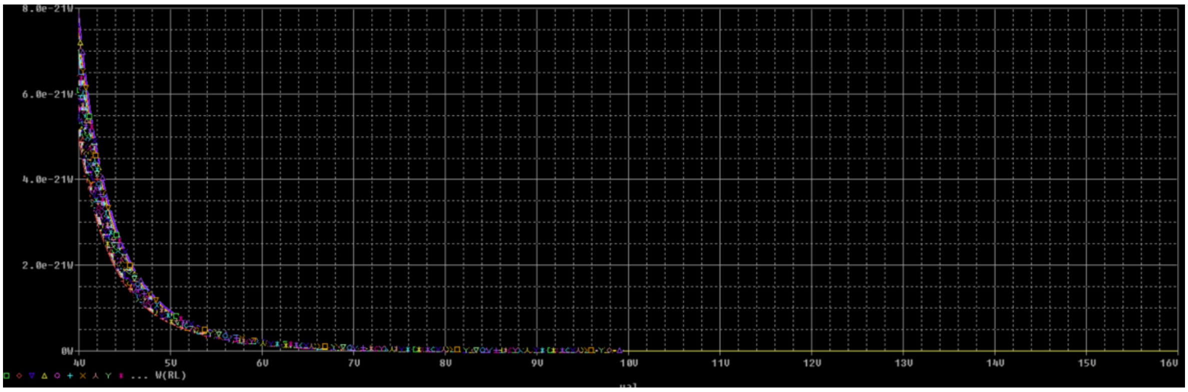


Image 16: Dissipated power by RL

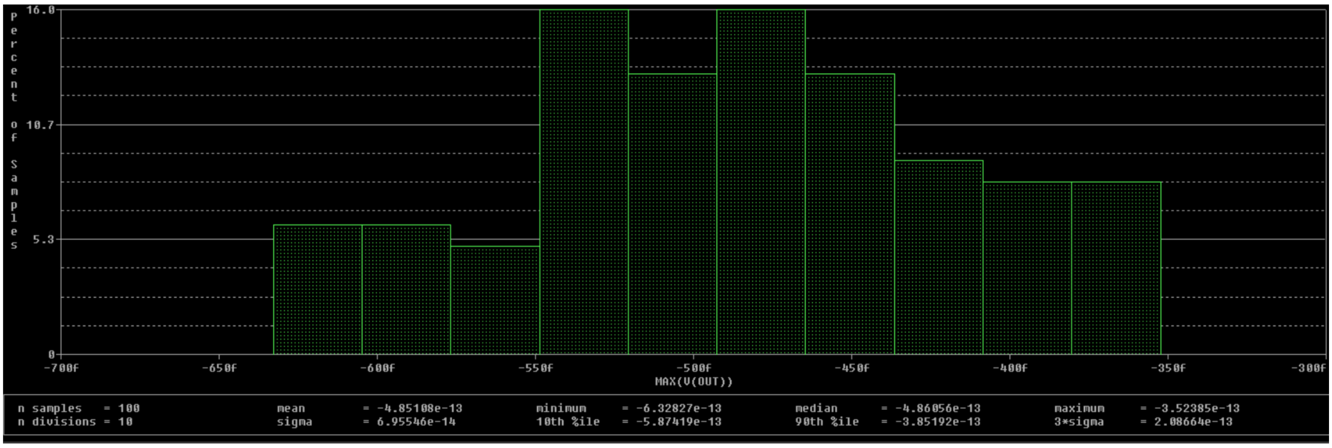


Image 17: MAX(V(out))

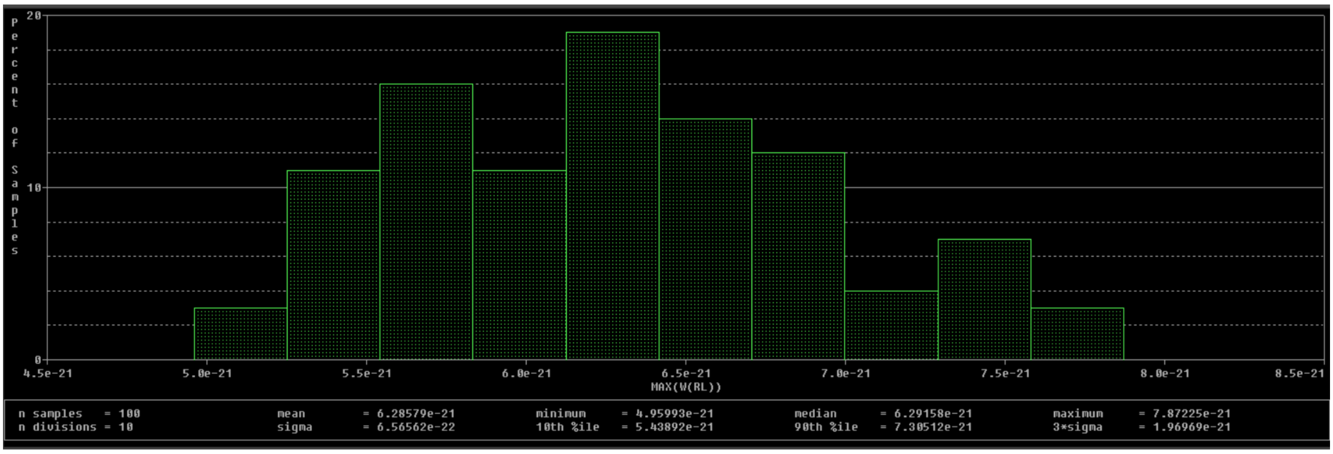


Image 18: MAX(W(RL))

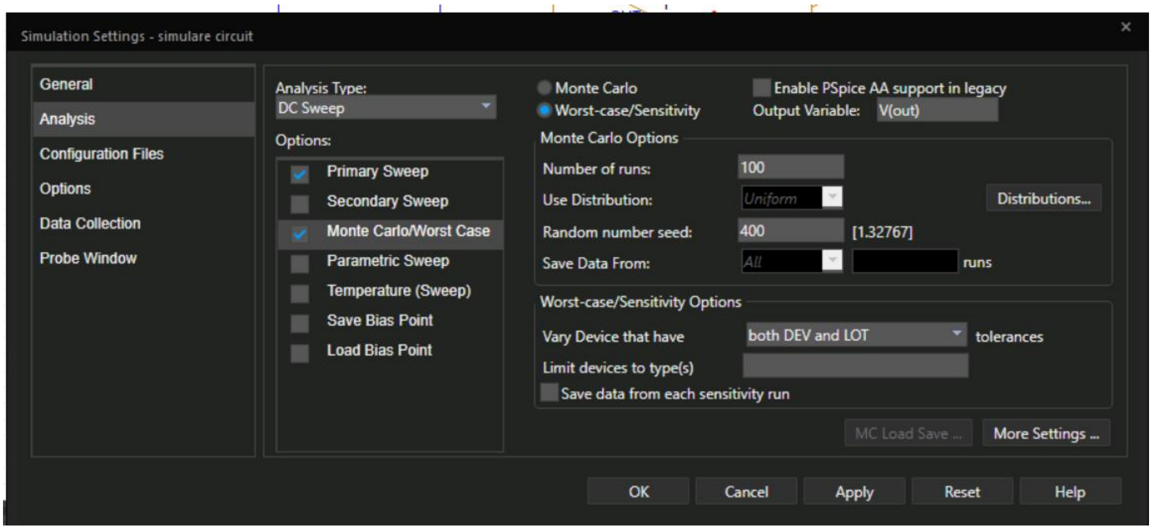


Image 19: WORST CASE settings

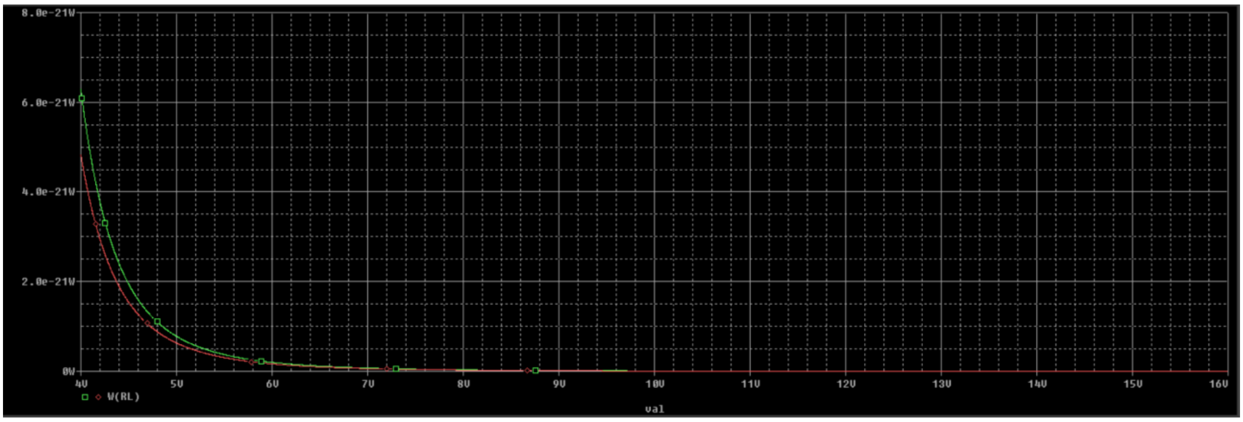
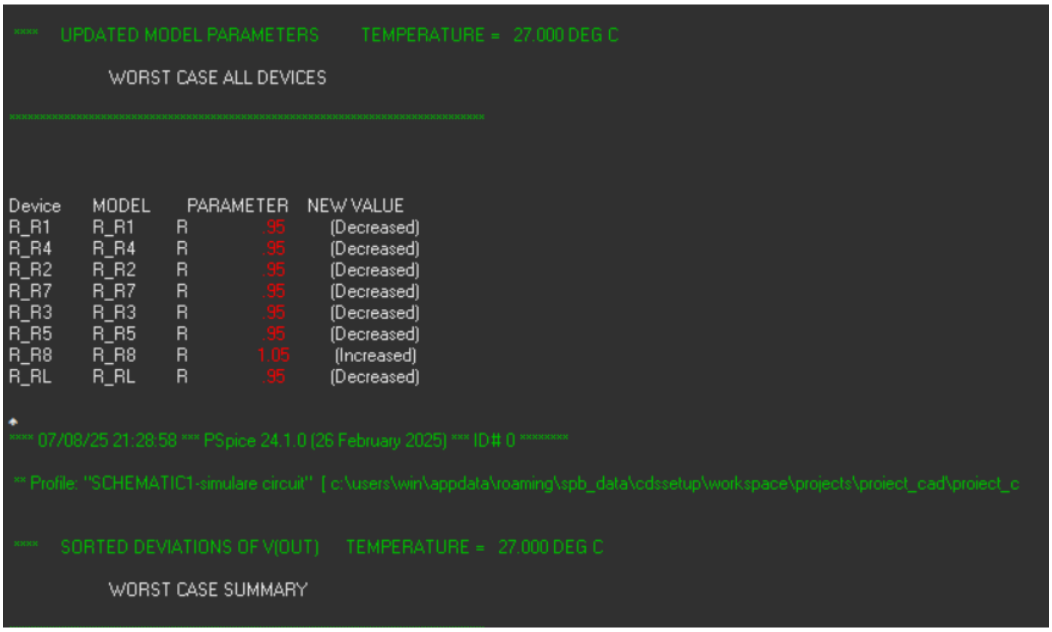


Image 20: Dissipated power by RL



**** UPDATED MODEL PARAMETERS TEMPERATURE = 27.000 DEG C

WORST CASE ALL DEVICES

Device	MODEL	PARAMETER	NEW VALUE	
R_R1	R_R1	R	.95	(Decreased)
R_R4	R_R4	R	.95	(Decreased)
R_R2	R_R2	R	.95	(Decreased)
R_R7	R_R7	R	.95	(Decreased)
R_R3	R_R3	R	.95	(Decreased)
R_R5	R_R5	R	.95	(Decreased)
R_R8	R_R8	R	1.05	(Increased)
R_RL	R_RL	R	.95	(Decreased)

**** 07/08/25 21:28:58 *** PSpice 24.1.0 (26 February 2025) *** ID# 0 ********

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**** SORTED DEVIATIONS OF V(OUT) TEMPERATURE = 27.000 DEG C

WORST CASE SUMMARY

Image 21: Analysis results

5. Conclusion

A voltage regulator is an electronic circuit which maintains the output voltage constant despite reasonable changes in the input voltage, load, temperature, etc. Voltage regulators keep the voltages from a power supply within a range that is compatible with the other electrical components. While voltage regulators are most commonly used for DC/DC power conversion, some can perform AC/AC or AC/DC power conversion as well. There are two main types of voltage regulators: linear and switching. Both types regulate a system's voltage, but linear regulators operate with low efficiency and switching regulators operate with high efficiency. In high-efficiency switching regulators, most of the input power is transferred to the output without dissipation.

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